

An alternating binomial sum of logarithms

a high finite difference is a tiny integral

$$S_n = \sum_{k=1}^{n-1} (-1)^k \binom{n}{k} \log(k+1) = (-1)^{n+1} \log(n+1) - I_n, \quad I_n = \int_0^1 \frac{(1-x)^n}{-\log x} dx,$$

$$I_n = \frac{1}{n \log n} \left(1 - \frac{\gamma}{\log n} + O\left(\frac{1}{\log^2 n}\right) \right), \quad 0 < I_n < \frac{1}{n}.$$

Overview. The alternating binomial coefficients are the n -th finite difference operator in disguise:

$$\Delta^n a_0 = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} a_j.$$

Applying this to $a_m = \log(m+1)$, the *completed* sum $T_n = \sum_{k=0}^n (-1)^k \binom{n}{k} \log(k+1)$ is exactly $(-1)^n \Delta^n a_0$. High-order differences of a slowly growing smooth function are small, essentially because $\log$ is nearly a polynomial of low degree on any short scale. Here the smallness is quantified exactly: $T_n = -I_n$ with I_n an explicit positive integral of size $\frac{1}{n \log n}$.

The sum S_n of the problem omits the terms $k=0$ (which is $\log 1 = 0$) and $k=n$ (which is $(-1)^n \log(n+1)$), so

$$S_n = T_n - (-1)^n \log(n+1) = (-1)^{n+1} \log(n+1) - I_n.$$

Thus S_n is dominated by the single omitted term $\pm \log(n+1)$, corrected by a quantity that vanishes like $\frac{1}{n \log n}$.

Two independent derivations of I_n are given: a discrete one, through the finite-difference calculus of $\frac{1}{m+\alpha}$, and a continuous one, through the Frullani-type representation of $\log(k+1)$. The first gives the sharp elementary bound $I_n < \frac{1}{n}$; the second explains why Euler's constant appears in the second-order term.

1. The sum is a finite difference

Let $a_m = \log(m+1)$ and $\Delta a_m = a_{m+1} - a_m$. Iterating and using Pascal's rule $\binom{r+1}{j} = \binom{r}{j} + \binom{r}{j-1}$,

$$\Delta^r a_m = \sum_{j=0}^r (-1)^{r-j} \binom{r}{j} a_{m+j}. \quad (1)$$

Setting $m=0$, $r=n$ and multiplying by $(-1)^n$,

$$T_n := \sum_{k=0}^n (-1)^k \binom{n}{k} \log(k+1) = (-1)^n \Delta^n a_0. \quad (2)$$

The sum of the problem omits the two extreme terms:

$$k=0 : \log 1 = 0, \quad k=n : (-1)^n \log(n+1),$$

so

$$S_n = T_n - (-1)^n \log(n+1). \quad (3)$$

Everything therefore reduces to showing that T_n is small, i.e. that the n -th finite difference of $\log(m+1)$ at $m=0$ is small.

2. Discrete evaluation: the harmonic ruler

Write the first difference as an average:

$$\Delta a_m = \log \left(1 + \frac{1}{m+1} \right) = \int_0^1 \frac{dt}{m+1+t}. \quad (4)$$

This is the key step: it expresses Δa_m as a superposition of the elementary sequences $m \mapsto \frac{1}{m+\alpha}$, for which the finite differences are known exactly:

$$(-1)^r \Delta^r \frac{1}{m+\alpha} \Big|_{m=0} = \frac{r!}{\alpha(\alpha+1) \cdots (\alpha+r)}, \quad \alpha > 0. \quad (5)$$

(Immediate by induction, since $\frac{1}{m+\alpha} - \frac{1}{m+1+\alpha} = \frac{1}{(m+\alpha)(m+1+\alpha)}$.)

Applying Δ^{n-1} to (4) and using (5) with $\alpha = 1+t$, $r = n-1$,

$$I_n := (-1)^{n-1} \Delta^n a_0 = \int_0^1 \frac{(n-1)! dt}{(1+t)(2+t) \cdots (n+t)}. \quad (6)$$

By (2), $T_n = -I_n$, which with (3) gives the first boxed formula.

Since the integrand of (6) is positive and, for $t \in [0, 1]$, bounded by its value at $t = 0$,

$$\boxed{0 < I_n < \frac{(n-1)!}{1 \cdot 2 \cdots n} = \frac{1}{n}.} \quad (7)$$

The harmonic ruler. Bound (7) is best understood by comparison with the harmonic numbers. Let $H_0 = 0$ and $H_m = 1 + \frac{1}{2} + \cdots + \frac{1}{m}$, so that $\Delta H_m = \frac{1}{m+1}$. By (5) with $\alpha = 1$,

$$(-1)^{n-1} \Delta^n H_0 = \frac{1}{n}. \quad (8)$$

Comparing (4) with $\Delta H_m = \frac{1}{m+1}$: the logarithm's first difference is the *average* of $\frac{1}{m+1+t}$ over $t \in [0, 1]$, hence strictly smaller than $\frac{1}{m+1}$. Since all the relevant differences carry the same alternating sign, the same ordering persists at order n , which is exactly (7). In words: $\log(m+1)$ and H_m have the same alternating-difference structure, and the logarithm sits below the harmonic ruler.

3. Continuous evaluation: a Frullani representation

An independent route gives a form better suited to asymptotics. For integers $k \geq 0$,

$$\log(k+1) = \int_0^1 \frac{1-x^k}{-\log x} dx, \quad (9)$$

which is the Frullani integral $\int_0^1 \frac{x^b - x^a}{\log x} dx = \log \frac{b+1}{a+1}$ at $a = 0$, $b = k$.

Insert (9) into T_n and exchange the finite sum with the integral. Since $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$ for $n \geq 1$ and $\sum_{k=0}^n (-1)^k \binom{n}{k} x^k = (1-x)^n$,

$$T_n = \int_0^1 \frac{0 - (1-x)^n}{-\log x} dx = - \int_0^1 \frac{(1-x)^n}{-\log x} dx.$$

Hence

$$\boxed{I_n = \int_0^1 \frac{(1-x)^n}{-\log x} dx = \int_0^1 \frac{(n-1)! dt}{(1+t)(2+t) \cdots (n+t)},} \quad (10)$$

the two representations agreeing, as one checks numerically to full precision for any n .

4. Asymptotics: where γ comes from

Use the integral representation. Substituting $x = \frac{y}{n}$ and writing $L = \log n$,

$$I_n = \frac{1}{n} \int_0^n \frac{(1 - y/n)^n}{L - \log y} dy. \quad (11)$$

Two expansions now govern the answer:

$$(1 - y/n)^n \rightarrow e^{-y}, \quad \frac{1}{L - \log y} = \frac{1}{L} \left(1 + \frac{\log y}{L} + O\left(\frac{\log^2 y}{L^2}\right) \right). \quad (12)$$

Therefore

$$I_n = \frac{1}{nL} \left(\int_0^\infty e^{-y} dy + \frac{1}{L} \int_0^\infty e^{-y} \log y dy + O\left(\frac{1}{L^2}\right) \right).$$

The two integrals are classical:

$$\int_0^\infty e^{-y} dy = 1, \quad \int_0^\infty e^{-y} \log y dy = \Gamma'(1) = -\gamma. \quad (13)$$

Hence

$$I_n = \frac{1}{n \log n} \left(1 - \frac{\gamma}{\log n} + O\left(\frac{1}{\log^2 n}\right) \right). \quad (14)$$

The same constant, discretely. Formula (6) reproduces (14) without any Gamma function. Factoring out $(n-1)!$,

$$I_n = \frac{1}{n} \int_0^1 \prod_{r=1}^n \left(1 + \frac{t}{r} \right)^{-1} dt = \frac{1}{n} \int_0^1 \exp \left(- \sum_{r=1}^n \log \left(1 + \frac{t}{r} \right) \right) dt, \quad (15)$$

and for $0 \leq t \leq 1$,

$$\sum_{r=1}^n \log \left(1 + \frac{t}{r} \right) = t H_n + O(t^2).$$

The integrand is therefore concentrated on $t \asymp H_n^{-1}$; substituting $t = u/H_n$,

$$I_n = \frac{1}{n H_n} \int_0^{H_n} e^{-u} \exp \left(O\left(\frac{u^2}{H_n^2}\right) \right) du = \frac{1}{n H_n} \left(1 + O\left(\frac{1}{H_n^2}\right) \right). \quad (16)$$

Expanding $H_n = \log n + \gamma + O(n^{-1})$ turns (16) into (14). So the γ in the second-order term is, on the discrete side, nothing but $H_n - \log n \rightarrow \gamma$, and on the continuous side the value $\Gamma'(1)$. The two occurrences are the same constant seen from the two ends of (10).

5. Numerical check

n	I_n (exact)	$\frac{1}{n \log n} \left(1 - \frac{\gamma}{\log n} \right)$	$\frac{1}{n H_n}$	bound $\frac{1}{n}$
5	0.0871265	0.0796993	0.0875912	0.2
20	0.0145850	0.0134745	0.0138976	0.05
100	0.00201125	0.00189930	0.00192776	0.01
1000	0.000137012	0.000132668	0.000133592	0.001

The discrete approximation $\frac{1}{n H_n}$ of (16) is visibly better than the two-term logarithmic expansion (14), which is expected: the expansion parameter of (14) is $\frac{1}{\log n}$, still 0.14 at $n = 1000$, whereas (16) resums the whole harmonic number.

6. Summary of the identity

Collecting (3), (10) and (14),

$$S_n = \sum_{k=1}^{n-1} (-1)^k \binom{n}{k} \log(k+1) = (-1)^{n+1} \log(n+1) - \frac{1}{n \log n} \left(1 - \frac{\gamma}{\log n} + O\left(\frac{1}{\log^2 n}\right) \right).$$

The sum is not small; the completed sum is. S_n grows like $\log n$ in absolute value and alternates in sign with n , whereas $T_n = S_n + (-1)^n \log(n+1)$ tends to 0. Completing the sum by the term $k = n$ is what turns it into a finite difference; the binomial coefficients only form a difference operator when the full range is present.

The cancellation is enormous. The individual terms of T_n reach size $\binom{n}{n/2} \log n \approx 2^n$, while the total is $\frac{1}{n \log n}$. Formula (10) explains the mechanism: the alternating sum of x^k with binomial weights is $(1-x)^n$, which is exponentially small everywhere except near $x = 0$, where the factor $\frac{1}{-\log x}$ is itself small. The competition between these two factors, resolved in Section 4, produces the scale $\frac{1}{n \log n}$: an algebraic $\frac{1}{n}$ from the width of the region where $(1-x)^n$ survives, and a logarithmic $\frac{1}{\log n}$ from the weight there.